Genuine Multipartite Entanglement Analysis

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Model                              Ground-state properties           GME summary
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Hubbard ring (1-D, PBC)            E0                 : -39.26003999 Extracted n-spin RDM : n = 3
                                   E1                 : -39.25394536
System parameters                                                    Purity:
-----------------                  <S2>               : 0.000000       gamma = Tr(rho^2)
Sites (L)          : 4             S                  : 0.0000
Electrons (Ne)     : 4             <Sz>               : 0.000000     Separable cut criterion:
Hopping (t)        : 10.0          Multiplicity 2S+1  : 1.0            gamma_A = 1  <->  rho = rho_A x rho_B
Interaction (U)    : 1.0           Spin state         : Singlet (S=0)
Mode               : uniform                                        C_GME (Troiani et al.):
n-body order       : 3             Double occupancy                   C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))
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                                   <D> total          : 0.730035     Decision logic:
                                   <D> / site         : 0.182509       gamma=1, all Tr(rho_A^2)<1  -> GME
                                                                       gamma=1, some Tr(rho_A^2)=1 -> No GME
                                                                       gamma < 1                   -> Mixed

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